

EXAMINING THE EFFECT OF DECLINING HIGH SCHOOL GRADUATES ON FIRST-YEAR ENROLLMENTS AT UIC

In Fall 2023, we started exploring multiple websites for publicly available High School Enrollment data. To address concerns around the imminent high school graduates' cliff around Fall 2025, we wanted to determine the effect of the enrollment cliff on the future Undergrad Enrollments at UIC.

This document goes through the exploratory data analysis conducted on the High School Enrollment dataset available on the Illinois State Board of Education (ISBE): <https://www.isbe.net/Pages/Illinois-State-Report-Card-Data.aspx>, and historical first-year enrollments at UIC.

The following files were utilized for the analysis in this document:

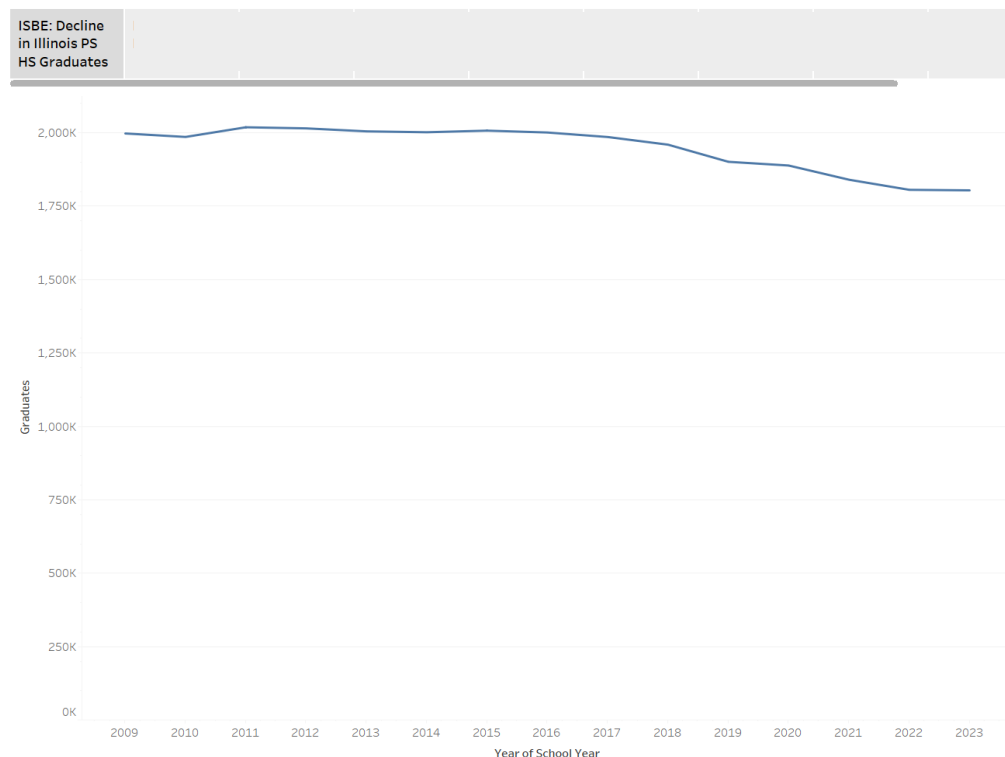
1. [Report Card Data from ISBE](#)
2. [UIC First Year Enrollments Data from EDW](#)

To examine interactively the results interpreted in this document, use the following workbook:

1. [Data Exploration](#)

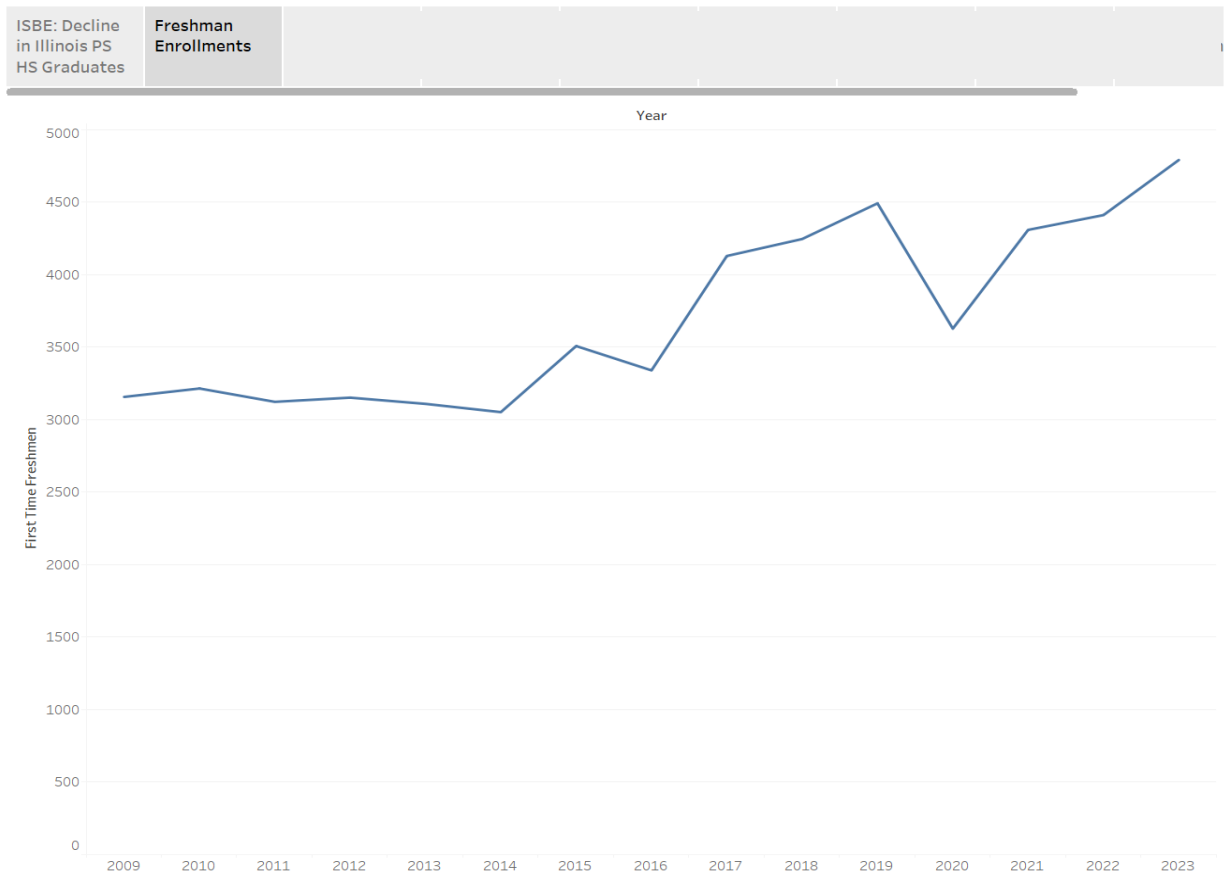
Trend of Illinois Secondary Education Graduates\

As seen in the below image, we observed a declining pattern in the high school graduates at Illinois Public Schools from 2009 until 2023, the date range during which the data was available in the ISBE Report Card data.



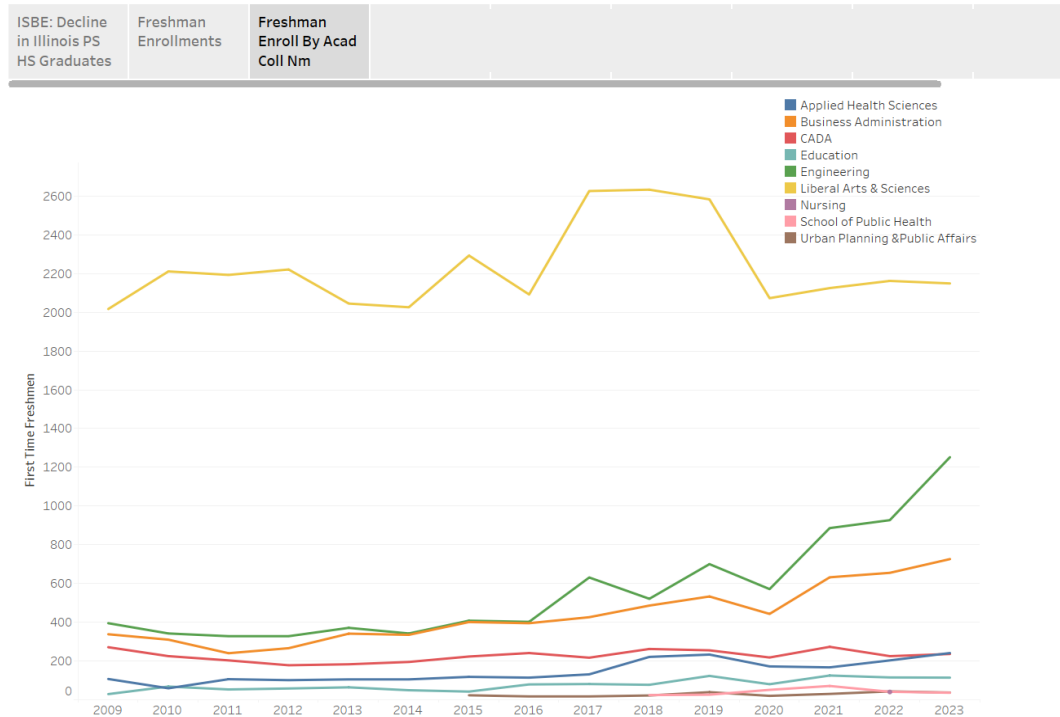
Trend of UIC Freshman Enrollments:

We observed an overall increasing pattern of UIC Freshman Enrollments, considering the first-year enrollments from 2009 until 2023.

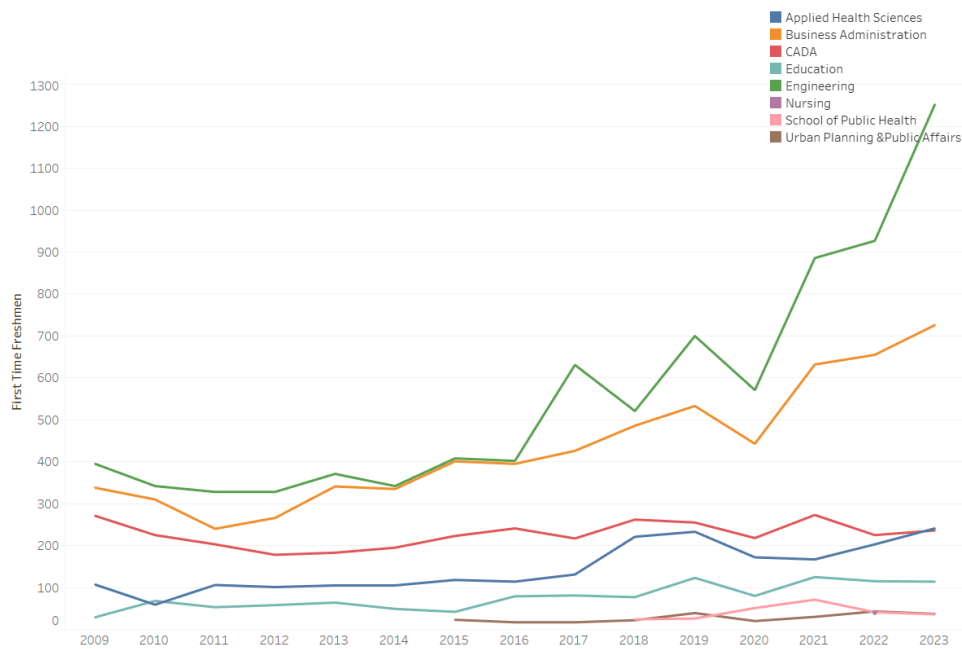


Trend of UIC Freshman Enrollments broken down by Academic Colleges:

Except for Liberal Arts and Sciences, a strong increasing pattern of UIC Freshman Enrollments was seen in the years from 2009 until 2023.

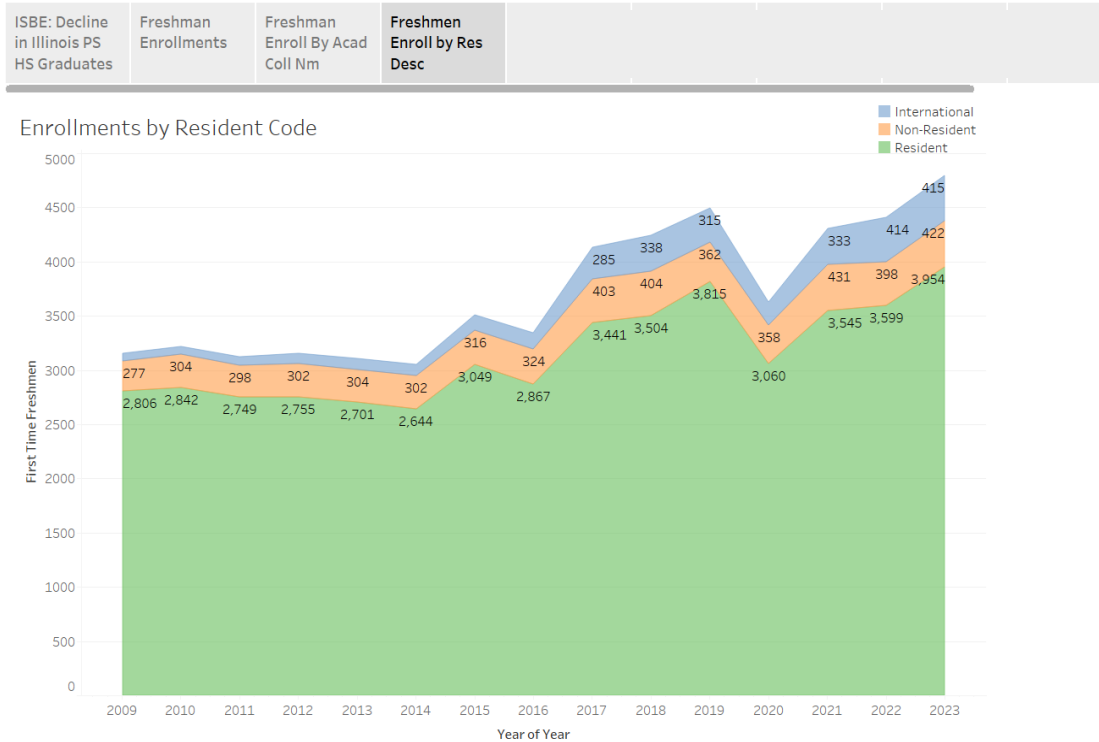


Here is the trend of the first-year enrollments at the rest of the colleges, with LAS filtered out from the results.

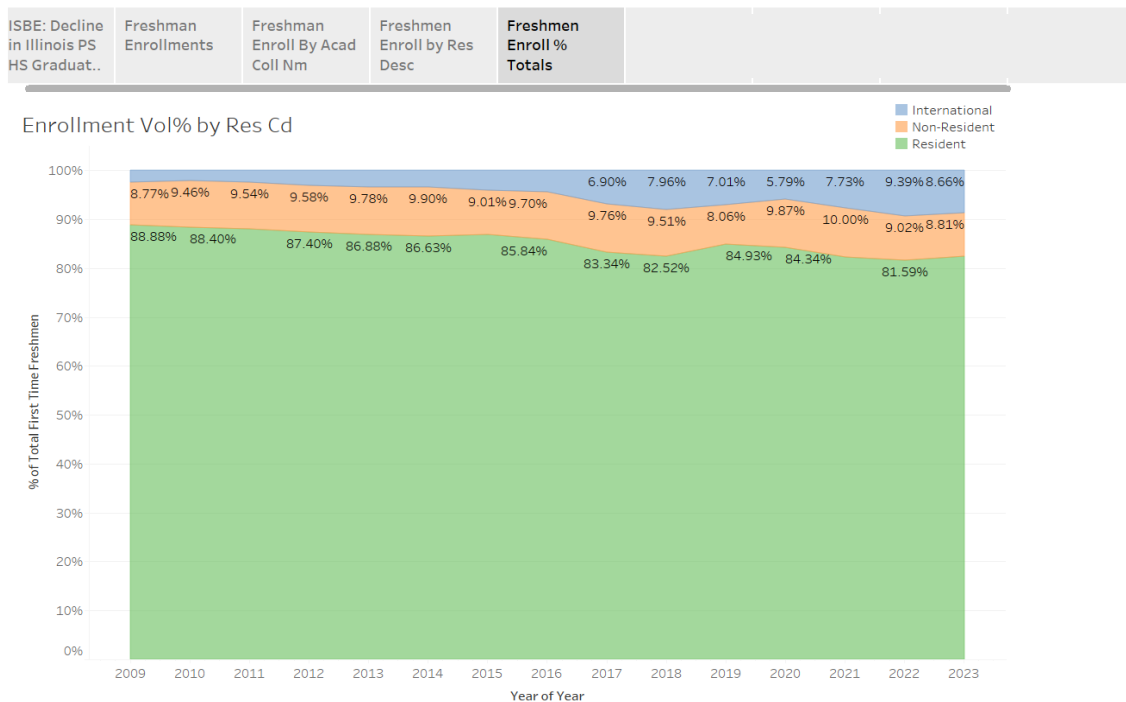


UIC Freshmen Enrollments broken down by Student Residence Status:

At this point, we conducted an analysis of the first-year enrollment numbers of UIC Campus broken down by the student residence status (Resident, Non-Resident, International)



There seemed to be a pattern emerging in terms of the enrollments broken down by residence status, and we explored further to examine the percentage totals of the three statuses in the UIC Freshmen Enrollment numbers.



We noticed the following pattern in the percentage totals of the three statuses:

1. % Total of Students Resident Status declined from 88.89% in 2009 to 82.53% in 2023.
2. % Total of Students Non-Resident Status has remained almost constant from 8.77% in 2009 to 8.66% in 2023.
3. % Total of Students International Status increased from 2.34% in 2009 to 8.66% in 2023.

A further correlation analysis was conducted between UIC First-year enrollments and ISBE's High School Graduates datasets at each college level:

UIC Enrollments vs HS Graduates (ISBE)		
Acad Coll Name		
Applied Health Sciences		-78.1%
Architecture,Design,& the Arts		-9.9%
Business Administration		-92.7%
Education		-83.9%
Engineering		-92.4%
Liberal Arts & Sciences		14.1%
School of Public Health		-40.5%
Urban Planning &Public Affairs		-75.9%

We made a preliminary conclusion that the disparity between ISBE Report Card's High School Graduates decline, and the increase in overall UIC Freshmen Enrollments is caused by the increase in the International Student intake at UIC between 2009 and 2023. Due to this, we determined that the available ISBE report card data does not suffice for incorporating High School Graduates data to improve our forecast on UIC Enrollments. We will need additional files that have information/data around International High School Graduates, and National Level High School Graduates. Without supporting files, ISBE report card data will not convey the necessary high school graduate information to the forecasting model.

Appendix:

1. [Python Script conducting the above correlation results](#)